2.57 Constant Acceleration: Vectors

Worked Solutions

1)

$\overrightarrow{AB} = \binom{8}{11} - \binom{-2}{4} = \binom{10}{7}$

$|\overrightarrow{AB}| = \sqrt{10^2 + 7^2} = \sqrt{149}$ m $= 12.2$ m (3 SF)

2 a)

$\sqrt{2^2 + 4^2} = 2\sqrt{5} = 4.47$ ms^{-1} (3 SF)

2 b)

$\sqrt{5^2 + 8^2} = \sqrt{89} = 9.43$ ms^{-1} (3 SF)

2 c)

$\sqrt{(2.5k)^2 + (6k)^2} = 6.5k$ ms^{-1} (3 SF)

3 a)

$\mathbf{a} = -2\mathbf{i} + 3\mathbf{j}$

$\sqrt{2^2 + 3^2} = \sqrt{13} = 3.61$ ms^{-2} (3 SF)

3 b)

$\mathbf{s} = -400\mathbf{i} + 250\mathbf{j}$

$\sqrt{400^2 + 250^2} = 50\sqrt{89} = 472$ m (3 SF)

3 c)

$\mathbf{v} = 12\mathbf{i} + 10\mathbf{j}$

$\sqrt{12^2 + 10^2} = 2\sqrt{61} = 15.6$ ms^{-1}

4)

$\mathbf{a} = 4\mathbf{i} + 7\mathbf{j}, \mathbf{u} = \mathbf{i} + 2\mathbf{j}, t = 3$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\mathbf{v} = (\mathbf{i} + 2\mathbf{j}) + (4\mathbf{i} + 7\mathbf{j}) \times 3 = (13\mathbf{i} + 23\mathbf{j})$ ms^{-1}

5 a)

$\mathbf{a} = -\mathbf{i} + 3\mathbf{j}, \mathbf{v} = -2\mathbf{i} + 5\mathbf{j}, t = 7$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$(-2\mathbf{i} + 5\mathbf{j}) = \mathbf{u} + (-\mathbf{i} + 3\mathbf{j}) \times 7$

$\mathbf{u} = (5\mathbf{i} - 16\mathbf{j})$ ms^{-1}

5 b)

$\sqrt{5^2 + 16^2} = \sqrt{281} = 16.8$ ms^{-2} (3 SF)

6)

$\mathbf{u} = \binom{4}{3}, t = 7, \mathbf{v} = \binom{8}{18}$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\binom{8}{18} = \binom{4}{3} + 7\mathbf{a}$ so $\mathbf{a} = \binom{\frac{4}{7}}{\frac{15}{7}}$

$a = \sqrt{\left(\frac{4}{7}\right)^2 + \left(\frac{15}{7}\right)^2} = 2.22 \text{ ms}^{-2}$ (3 SF)

7 a)

$\binom{10}{0}$ ms^{-1}

7 b)

$\sqrt{1^2 + 1^2} = \sqrt{2}$

$4 \times \frac{1}{\sqrt{2}}\binom{1}{1} = \frac{1}{\sqrt{2}}\binom{4}{4}$ ms^{-1}

7 c)

$\sqrt{2^2 + 3^2} = \sqrt{13}$

$0.3 \times \frac{1}{\sqrt{13}}\binom{2}{-3} = \frac{3\sqrt{13}}{130}\binom{2}{-3}$ ms^{-1}

8)

At $t = 2, \mathbf{r} = 2^2\mathbf{i} + (3 - 2 \times 2)\mathbf{j} = 4\mathbf{i} - \mathbf{j}$

At $t = 5, \mathbf{r} = 5^2\mathbf{i} + (3 - 2 \times 5)\mathbf{j} = 25\mathbf{i} - 7\mathbf{j}$

$(25\mathbf{i} - 7\mathbf{j}) - (4\mathbf{i} - \mathbf{j}) = 21\mathbf{i} - 6\mathbf{j}$ metres

9 a)

$\mathbf{u} = (2\mathbf{i} + 5\mathbf{j}), \mathbf{a} = (6\mathbf{i} - \mathbf{j})$

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

$\mathbf{s} = (2\mathbf{i} - 5\mathbf{j})t + \frac{1}{2}(6\mathbf{i} - \mathbf{j})t^2$

$= (2t + 3t^2)\mathbf{i} + \left(5t - \frac{1}{2}t^2\right)\mathbf{j}$

9 b)

$\mathbf{s} = (2 \times 10 + 3 \times 10^2)\mathbf{i} + \left(5 \times 10 - \frac{1}{2} \times 10^2\right)\mathbf{j}$

$= 320\mathbf{i} + 0\mathbf{j}$

320 metres

10 a)

$\mathbf{a} = 5\mathbf{i} + 4\mathbf{j}, \mathbf{u} = 0$

$\mathbf{s} = (2\mathbf{i} - 7\mathbf{j}) + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

$\mathbf{s} = (2\mathbf{i} - 7\mathbf{j}) + \frac{1}{2}(5\mathbf{i} + 4\mathbf{j})t^2$

$\mathbf{s} = \left(2 + \frac{5}{2}t^2\right)\mathbf{i} + (2t^2 - 7)\mathbf{j}$

10 b)

$\mathbf{r} = 12\mathbf{i} + \mathbf{j}$

$\sqrt{12^2 + 1^2} = \sqrt{145} = 12.0$ metres (3 SF)

11)

$\mathbf{u} = \mathbf{i} - 3\mathbf{j}, \mathbf{a} = 8\mathbf{i} - 5\mathbf{j}, t = 4$

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

From initial position:

$\mathbf{s} = (\mathbf{i} - 3\mathbf{j}) \times 4 + \frac{1}{2}(8\mathbf{i} - 5\mathbf{j}) \times 4^2 = 68\mathbf{i} - 52\mathbf{j}$

$\mathbf{s}_0 = (6\mathbf{i} + 2\mathbf{j})$

So $6\mathbf{i} + 2\mathbf{j} + 68\mathbf{i} - 52\mathbf{j} = 74\mathbf{i} - 50\mathbf{j}$ metres

12)

$\mathbf{u} = \binom{-1}{3}, \mathbf{a} = \binom{2}{-1}, t = 4$

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

From initial position:

$\mathbf{s} = \binom{-1}{3} \times 4 + \frac{1}{2}\binom{2}{-1} \times 4^2 = \binom{12}{4}$

$\mathbf{s}_A + \binom{12}{4} = \binom{8}{-2}$

$\mathbf{s}_A = \binom{-4}{-6}$ metres

13 a)

$\mathbf{a} = (2\mathbf{i} - 4\mathbf{j}), \mathbf{v} = (4\mathbf{i} + 3\mathbf{j}), t = T, \mathbf{s} = (4\mathbf{i} + 3\mathbf{j})$

$\mathbf{s} = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2$

$(4\mathbf{i} + 3\mathbf{j}) = (4\mathbf{i} + 3\mathbf{j})T - \frac{1}{2}(2\mathbf{i} - 4\mathbf{j})T^2$

$-12 = 4T - T^2$

Solving $T^2 - 4T - 12 = 0$ gives $T = 6$ or $T = -2$ so $T = 6$

13 b)

$k = 3 \times 6 - \frac{1}{2} \times -4 \times 6^2 = 90$

14)

$\binom{k}{78} - \binom{5}{-2} = \binom{k-5}{80}$

$\frac{80}{k-5} = \frac{10}{7}$

$560 = 10(k - 5) \Rightarrow k = 61$

15 a)

$k = 8$

15 b)

$k = -6$

15 c)

$k = -9$

15 d)

Any value of k

16)

$\mathbf{a} = \binom{2}{3}, \mathbf{u} = \binom{9}{2}, t = T$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\mathbf{v} = \binom{9}{2} + \binom{2}{3}T = \binom{9+2T}{2+3T}$

$\binom{9+2T}{2+3T}$ is parallel to $\binom{4}{1}$

$\frac{2+3T}{9+2T} = \frac{1}{4}$

$\Rightarrow 4(2+3T) = 9+2T \Rightarrow 8+12T = 9+2T$

$\Rightarrow 10T = 1 \Rightarrow T = \frac{1}{10} = 0.1$ seconds

17 a)

$2(4.2\mathbf{i} + 14.4\mathbf{j}) = 8.4\mathbf{i} + 28.8\mathbf{j}$ ms^{-1}

17 b)

Emily's speed: $\sqrt{4.2^2 + 14.4^2} = 15$ ms^{-1}

$\frac{1}{3}(4.2\mathbf{i} + 14.4\mathbf{j}) = 1.4\mathbf{i} + 4.8\mathbf{j}$ ms^{-1}

17 c)

Emily's speed: $\sqrt{4.2^2 + 14.4^2} = 15$ ms^{-1}

$\frac{4}{5}(4.2\mathbf{i} + 14.4\mathbf{j}) = 3.36\mathbf{i} + 11.52\mathbf{j}$ ms^{-1}

17 d)

Emily's speed: $\sqrt{4.2^2 + 14.4^2} = 15$ ms^{-1}

$-\frac{1}{2}(4.2\mathbf{i} + 14.4\mathbf{j}) = -2.1\mathbf{i} - 7.2\mathbf{j}$ ms^{-1}

18 a)

$\sqrt{9^2 + 21.6^2} = 23.4$ ms^{-1}

18 b)

Particle B:

$13 \times \frac{1}{23.4}\binom{9}{21.6} = \binom{7.5}{18}$ ms^{-1}

18 c)

$\binom{2}{-3} + 5\binom{7.5}{18} = \binom{39.5}{87}$ metres

19)

$\sqrt{4^2 - \left(\frac{20-16}{2}\right)^2} = 2\sqrt{3} = 3.46$ metres (3 SF)

20 a)

$\sqrt{2.4^2 + 5.76^2} = 6.24$ ms^{-1}

20 b)

$\mathbf{v} = 5.2 \times \frac{1}{6.24}(2.4\mathbf{i} + 5.76\mathbf{j}) = 2\mathbf{i} + 4.8\mathbf{j}$

$\mathbf{s} = (-\mathbf{i} + 3\mathbf{j}) + 2.5(2\mathbf{i} + 4.8\mathbf{j})$ metres

So $(4\mathbf{i} + 15\mathbf{j})$ metres

20 c)

Vector from where the first squirrel starts to where it is after 2.5 seconds is $5\mathbf{i} + 12\mathbf{j}$

Vector from where the first squirrel starts to where the second squirrel starts is $12\mathbf{i} - 5\mathbf{j}$

These vectors are perpendicular. As the second squirrel is travelling faster than the first squirrel, they will never be closer than they are initially.

$\therefore \sqrt{12^2 + (-5)^2} = 13$ metres

21)

$\binom{-7}{-1} + t\binom{6}{3} = \binom{0}{2} + t\binom{2.5}{1.5}$

$-7 + 6t = 2.5t \Rightarrow t = 2$

$-1 + 3t = 2 + 1.5t \Rightarrow t = 2$

Same position at $t = 2$ $\therefore$ they collide.

22)

Particle A has speed $\sqrt{3.5^2 + 12^2} = 12.5$ ms^{-1}

$4 \times 12.5 = 50$ metres

Particle B has speed 6.25 ms^{-1}

$6 \times 6.25 = 37.5$ metres

$\sqrt{50^2 + 37.5^2} = 62.5$ m

23 a)

East

23 b)

South

23 c)

North-East

23 d)

North-West

24 a)

$\mathbf{u} = \binom{4}{0}, \mathbf{v} = \binom{0}{-2}, t = 40$

$\mathbf{s} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$

$\mathbf{s} = \frac{1}{2}\binom{4}{-2} \times 40 = \binom{80}{-40}$ m

24 b)

$\sqrt{80^2 + (-40)^2} = 40\sqrt{5} = 89.4$ m (3 SF)

25 a)

$t - 5 = 0 \Rightarrow t = 5$

$2t - 9 = 0 \Rightarrow t = \frac{13}{2}$

Not the same times, therefore, is never at rest.

25 b)

$t - 5 = 0 \Rightarrow t = 5$

$\mathbf{v} = -3\mathbf{j}$ which is due south so $t = 5$

25 c)

Equal positive components:

$t - 5 = 2t - 13 \Rightarrow t = 8$

26 a)

$t^2 = 0$ so at $t = 0$

26 b)

$105 + 8t = t^2$

Solving $t^2 - 8t - 105$ gives $t = 15$ and $t = -7$

So at $t = 15$

26 c)

Since $t \geq 0$

$105 + 8t \geq 105$ so cannot move West

27)

$\mathbf{u} = \binom{3}{8}, \mathbf{a} = \binom{2}{-20}$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\mathbf{v} = \binom{3}{8} + \binom{2}{-20}t = \binom{3+2t}{8-20t}$

$8 - 20t = 0$ so $t = 0.4$ seconds

28)

$\mathbf{u} = \binom{1}{3}, \mathbf{a} = \binom{15}{5}$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\mathbf{v} = \binom{1}{3} + \binom{15}{5}t = \binom{1+15t}{3+5t}$

$1 + 15t = 3 + 5t \Rightarrow 10t = 2$

so $t = 0.2$ seconds

29 a)

$\mathbf{u} = \binom{6}{-12}, \mathbf{v} = \binom{-2}{2}$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\binom{-2}{2} = \binom{6}{-12} + 2\mathbf{a} \Rightarrow \mathbf{a} = \binom{-4}{7}$ ms^{-1}

29 b)

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\mathbf{v} = \binom{6}{-12} + \binom{-4}{7}t = \binom{6-4t}{-12+7t}$

Due south:

$6 - 4t = 0$ so at $t = 1.5$ seconds

29 c)

$\binom{6-4\times1.5}{-12+7\times1.5} = \binom{0}{-1.5}$ so 1.5 ms^{-1} due south

30 a)

$\mathbf{a} = -\mathbf{i} - 3\mathbf{j}, \mathbf{u} = 13\mathbf{i} - 5\mathbf{j}, t = 5$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\mathbf{v} = (13\mathbf{i} - 5\mathbf{j}) + 5(-\mathbf{i} - 3\mathbf{j}) = 8\mathbf{i} - 20\mathbf{j}$

$\sqrt{8^2 + 20^2} = 4\sqrt{29} = 21.5$ ms^{-1} (3 SF)

30 b)

$\arctan\left(\frac{20}{8}\right) = 68.19859\ldots$

$90 + 68.19859\ldots$

Bearing of 158°

31 a)

$\mathbf{a} = 0.4\mathbf{i} + 0.2\mathbf{j}, \mathbf{u} = 0, t = 20$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\mathbf{v} = \mathbf{0} + 15(0.4\mathbf{i} + 0.2\mathbf{j}) = 6\mathbf{i} + 3\mathbf{j}$

$\sqrt{6^2 + 3^2} = 3\sqrt{5} = 6.71$ ms^{-1} (3 SF)

31 b)

$\arctan\left(\frac{3}{6}\right) = 26.56505118\ldots$

$90 - 26.56505\ldots = 63.43494882\ldots$

Bearing of 063°

31 c)

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

$\mathbf{s} = \mathbf{0} + \frac{1}{2}(0.4\mathbf{i} + 0.2\mathbf{j})t^2 = (0.2t^2)\mathbf{i} + (0.1t^2)\mathbf{j}$

$240 = \sqrt{(0.2t^2)^2 + (0.1t^2)^2}$

$57600 = 0.04t^4 + 0.01t^4$

$57600 = 0.05t^4$

$t = 32.8$ seconds (3 SF)